

QUMARLEB, China

WMO: 560210

Lat: 34.1287N

Lon: 95.8060E

Elev: 13806

StdP: 8.70

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-8.5	-32.2	1.5	3.4	-28.7	1.9	7.3	28.4	33.9	24.1	27.4	0.5	90	0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	65.9	46.9	63.3	46.5	60.8	45.6	50.1	61.6	48.8	59.5	47.6	57.5	7.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.6	76.9	54.9	44.5	73.6	53.5	43.4	70.5	52.5	26.1	61.9	25.1	60.1	24.2	57.5	54.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.9	18.4	15.6	DB	-19.0	71.0	3.0	2.9	-21.1	73.1	-22.9	74.8	-24.6	76.4	-26.8	78.5
(5)				WB	-19.8	52.6	2.9	1.0	-21.9	53.3	-23.6	53.9	-25.2	54.4	-27.3	55.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	10.1	15.4	22.4	31.4	38.9	45.8	50.5	49.8	43.7	31.8	19.7	12.3	
	DBStd	15.24	6.59	6.58	6.00	5.45	4.66	4.75	4.14	4.48	4.99	6.51	5.77	6.02	
	HDD50	7043	1235	970	857	558	344	139	45	60	194	563	908	1170	
	HDD65	12383	1700	1390	1322	1008	808	575	450	471	638	1028	1358	1635	
	CDD50	136	0	0	0	0	1	14	60	54	7	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	5.2	5.0	6.2	6.5	6.3	5.9	5.3	4.9	4.7	4.5	4.7	4.2	4.2	
Precipitation	PrecAvg	16.0	0.1	0.1	0.3	0.4	1.5	3.2	3.7	3.0	2.8	0.7	0.1	0.1	
	PrecMax	20.6	0.3	0.3	0.6	1.0	3.7	5.6	6.8	5.6	4.8	2.5	0.4	0.3	
	PrecMin	11.3	0.0	0.0	0.0	0.1	0.3	0.5	1.5	1.1	1.4	0.1	0.0	0.0	
	PrecStd	2.4	0.1	0.1	0.1	0.2	0.7	1.1	1.1	1.1	0.9	0.6	0.1	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	41.0	48.5	55.5	63.8	65.4	70.0	70.5	63.6	57.5	42.6	38.7	
		MCWB	23.0	24.5	28.6	33.2	39.4	45.6	49.2	46.7	45.6	38.2	26.5	23.3	
	2%	DB	33.1	36.3	43.0	51.2	57.6	61.6	65.9	66.2	60.0	50.7	39.2	33.7	
		MCWB	20.1	22.2	26.3	31.7	37.5	45.1	47.9	47.2	45.4	36.0	24.7	20.6	
	5%	DB	28.6	33.1	39.5	47.8	53.8	58.6	63.3	63.2	56.7	46.5	36.0	30.0	
		MCWB	17.6	20.4	24.9	30.3	36.1	44.0	47.1	47.0	43.8	33.5	23.3	18.6	
	10%	DB	24.6	30.0	36.3	44.5	50.4	55.8	60.6	60.2	53.5	43.2	32.8	26.6	
		MCWB	15.5	19.0	23.4	29.0	35.0	42.9	46.1	46.0	42.5	31.8	21.8	16.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	25.2	29.5	35.6	42.9	49.9	52.0	51.5	48.6	41.4	27.9	24.2
			MCDB	36.6	38.5	44.3	48.8	54.8	60.3	64.1	62.8	59.4	53.7	38.8	37.1
2%		WB	20.5	22.7	27.3	33.3	40.1	47.7	50.4	49.8	46.8	38.2	25.7	21.1	
		MCDB	32.0	34.7	40.9	47.0	51.4	57.7	62.3	61.1	57.1	47.2	36.6	32.6	
5%		WB	18.1	21.0	25.6	31.6	38.5	46.0	49.2	48.5	45.3	35.7	24.1	19.0	
		MCDB	27.8	32.2	38.1	44.5	49.1	55.2	60.2	59.4	54.5	43.9	34.6	29.4	
10%		WB	15.8	19.2	24.0	30.1	36.9	44.5	47.9	47.3	43.7	33.3	22.3	17.0	
		MCDB	24.1	29.2	35.3	41.9	47.0	53.2	57.7	57.4	51.5	40.3	31.9	26.1	
Mean Daily Temperature Range	5% DB	MDBR	27.2	26.0	25.0	23.0	20.1	17.7	18.7	19.4	18.6	20.4	25.7	28.4	
		MCDBR	32.3	30.7	30.3	28.5	27.2	22.2	23.4	24.6	23.1	24.5	29.4	32.8	
		MCWBR	22.5	20.3	17.7	13.8	12.0	9.5	9.3	9.6	10.5	11.7	17.4	22.5	
	5% WB	MCDBR	30.3	28.4	27.0	24.1	20.8	17.4	18.7	18.8	19.3	20.7	26.6	31.8	
		MCWBR	21.4	19.0	16.0	11.9	10.0	8.3	8.2	7.9	9.2	10.4	16.1	21.9	
Clear-Sky Solar Irradiance	taub	0.186	0.211	0.259	0.299	0.311	0.297	0.286	0.280	0.258	0.234	0.196	0.179		
	taud	2.612	2.540	2.399	2.293	2.342	2.496	2.591	2.599	2.627	2.617	2.635	2.674		
	Ebn at Noon	342	338	325	313	308	311	314	316	321	325	335	341		
	Edh at Noon	23	28	35	41	40	34	31	30	28	25	22	20		
All-Sky Solar Radiation	RadAvg	1047	1313	1685	1975	2001	1868	1917	1743	1528	1346	1197	1012		
	RadStd	95	67	88	119	135	124	123	131	95	69	43	60		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		+1.15	+2.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)		+0.83	+1.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page